Power Budget

Team Number:	202
Project Name:	**Temp/Hum Subsystem**
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Version:	**V1**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements

All Major Components	Component Name	Part Number	upplyVoltageRange	#	bsoluteMaximumCurrent (mA)	alCurrent(mA)	Unit
	Microcontroller	ESP32-S3-WROOM-1-I	3.0V–3.6 V	1	500	500	mA
	Humidity Sensor	HDC2080	1.62V–3.6 V	1	0.89	0.89	mA
	Op-Amp	MCP6001-OT	1.8V–6.0 V	1	0.17	0.17	mA
	Voltage Reference (2.5V out)	REF3325AIRSER	2.7V - 5.5V	1	0.005	0.005	mA
	3.3V buckregulator	XL1509-3.3	+5V - 35V	1	10	10	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.

+3.3V Power Rail	Component Name	Part Number	upplyVoltageRange	#	bsoluteMaximumCurrent (mA)	alCurrent(mA)	Unit
	Microcontroller	ESP32-S3-WROOM-1-I	3.0V–3.6 V	1	500	350	mA
	Humidity Sensor	HDC2080	1.62V–3.6 V	1	0.17	0.17	mA
	Op-Amp	MCP6001-OT	1.8V–6.0 V	1	0.005	0.005	mA
	Voltage Reference (2.5V out)	REF3325AIRSER	2.7V - 5.5V	1	10	10	mA
						360.175	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	450.21875	mA
c4. Regulator or Source Choice	+3.3V buck regulator	XL1509-3.3	+5V - 35V	1	500	500	mA
					Total Remaining Current Available on 3.3V Rail	49.78125	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple

External Power Source 1	Component Name	Part Number	upplyVoltageRange	Output	bsoluteMaximumCurrent (mA)	alCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+9V	5000	5000	mA
Power Rails Connected to External Power Source 1	+3.3V buck regulator	XL1509-3.3	+12V - 35V	1	10	10	mA
						0	mA
						0	mA
					Total Remaining Current Available on External Power Source 1	4990	mA

External Power Source 2	Component Name	Part Number	upplyVoltageRange	Output	bsoluteMaximumCurrent (mA)	alCurrent(mA)	Unit
Power Source 2 Selection	Battery	(full part number)	+9V	-9V	500	500	mA
Power Rails Connected to External Power Source 2	+3.3V buck regulator	XL1509-3.3	+5V - 35V	1	10	10	mA
					Total Remaining Current Available on External Power Source 2	490	mA

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.

	Component Name	Part Number	upplyVoltageRange	Capacity(mAh)	redByRegulators	
	Battery	(full part number)	+9V	500	10	
				Battery Life	50	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit